

YUGESHKARAN P

Full Stack AI Engineer · GenAI & Agentic Systems · LangChain · CrewAI · RAG · PyTorch
yugeshkaran01@gmail.com · +91 73975 92742 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#) · Bangalore, India

PROFESSIONAL SUMMARY

Full Stack AI Engineer with hands on experience in building and deploying production-grade agentic AI systems using Python, LangChain, CrewAI, and PyTorch. Hands-on expertise architecting multi-agent LLM workflows, RAG pipelines, and secure API backends with measurable production outcomes. Experienced in bridging AI prototypes to enterprise-grade deployments — implementing guardrails, rate limiting, JWT/OAuth authentication, Redis caching, and CORS security at the system level. Adept at co-building with technical teams and converting real-world AI integration challenges into scalable reusable solutions.

TECHNICAL SKILLS

Agentic AI & LLMs: LangChain, LangGraph, LangSmith, CrewAI, Sarvam, ReAct, multi-agent orchestration, MCP, tool-calling, Gemini

ML & Deep Learning: PyTorch, HuggingFace Transformers, NLP, RAG pipelines, embedding models, LLM inference optimization

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C — data structures, algorithms, software design

Backend & Security: Flask, Node.js, Express.js — REST APIs, JWT/OAuth, rate limiting, CORS, Redis caching, guardrails

Cloud & DevOps: AWS (S3, Lambda, Amplify), Docker, Git/GitHub, CI/CD, Linux/Unix, Vercel

Databases & Vectors: MySQL, MongoDB, Redis, Pinecone VectorDB — semantic retrieval, schema design, query optimization

WORK EXPERIENCE

Bytes Base — Founder [BytesBase.me](#) [LinkedIn Page](#)

Sep-2025 - Present

Multi-tenant AI assisted Developer Community Platform for Academic Institutions. Designed for students, alumni and faculties to collaborate and grow together. It aims to foster Tech culture inside the academic institutes.

- Architected and developed Bytes Base, using microservices architecture with Node.js and Flask services, enabling modular deployment and independent service scaling.
- Engineered server-side **Redis caching** for content feed APIs to minimize redundant data fetching, maximizing database throughput and API efficiency, used **JWT authentication** across all client requests, eliminating session-store performance bottlenecks.
- Engineered API-level **rate limiting middleware** preventing **100% of abusive requests and DoS attack vectors**, built real-time Socket.io discussions, AWS S3 media storage, and deployed on Vercel with CI/CD practices.
- Built responsive React.js/TailwindCSS frontend and Framer motion, deployed on Vercel with CI/CD practices and maintained high availability under concurrent user load.

INTERNSHIP EXPERIENCE

Full-Stack Web Development Intern · Quantum Sharq Innovative Solutions

2024

- Built and optimized React.js/TailwindCSS web applications for clients, developed modular HRMS components in an Agile team, maintaining code quality through reviews and testing.

Data Science Intern · Comp Allah IT Industry

2023

- Performed data analysis, ML model training, and Python diagnostic scripting on business datasets, built visualizations to drive data-informed decision-making workflows.

RESEARCH PUBLICATIONS

- (Jun 2025)** "Evaluating the Performance of LLMs on SQL and NoSQL Databases Using LangSmith" — Intl. Conf. Multidisciplinary Solutions for Emerging Trends 2025
- (Sep 2024)** "Design and Development of an Attendance Management and Analysis System Using LLM" — ICIAAIEA 2024

PROJECTS

Indic Research Scholar — Production RAG System [Github](#)

Flask · Sarvam Models · Pinecone VectorDB · Node.js · MongoDB · React.js · TailwindCSS

- Full Stack RAG platform helps researchers to query academic papers in 8+ Indian languages, with OCR ingestion and voice I/O via Sarvam AI (105B LLM, Saaras v3 STT, Bulbul v3 TTS).
- Built end-to-end auth (JWT + GitHub OAuth), multi-tenant document isolation, and a synchronous.

Assistant Knowledge Hub — Production RAG System [Github](#)

Python · Flask · OpenAI · Pinecone VectorDB · LangChain · Pydantic

- Built a production-grade RAG assistant with OpenAI text-embedding-3-small and Pinecone VectorDB, enforced **20 requests/minute/user rate limiting** to optimize LLM API costs and ensure equitable resource allocation across concurrent users.
- Implemented **CORS middleware blocking 100% of cross-site request forgery** attempts and deployed **LLM guardrails preventing 100% of prompt injection attacks** — delivering enterprise-grade security for a production agentic deployment.

Mini Language Model — Custom NLP Training [Github](#)

Python · PyTorch · Transformers · Flask

- Designed and trained a 2.96M-parameter transformer-based LLM from scratch using PyTorch.
- Built an end-to-end pipeline covering tokenization, architecture design, optimization, and inference speed and memory efficiency tuning and exposed via Flask inference API.

EDUCATION

B.Tech — Artificial Intelligence & Data Science · Dhanalakshmi Srinivasan University, Trichy

2021 – 2025

CGPA: **9.21 / 10.0** · Class XII (CBSE): 92%